



Transportation's Systems 20-5751

PROJECT PART 3: ONE-DIMENSIONAL OPTIMIZATION METHODS

PROF. ZAHRA AMINI
KEIVAN JAMALI

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1 Introduction

In this part, you will implement two fundamental one-dimensional optimization algorithms and apply them to transportation-related problems based on the Sioux Falls network.

The goal is to prepare you for more advanced algorithms (e.g., FrankWolfe) by first mastering line search and root-finding techniques.

Problem 1: Bisection Method for Flow Equilibrium on a Link

In transportation systems, congestion effects are often modeled using nonlinear travel time functions. Consider the BPR function:

$$T(f) = T^0 \left(1 + B \left(\frac{f}{C} \right)^4 \right)$$

We define the total cost on a link as:

$$Z(f) = f \cdot T(f)$$

Your task is to find the flow value f^* such that:

$$\frac{dZ(f)}{df} = K$$

Where:

- K is a constant derived from your student ID
- $C = 1000$
- T^0 and B are generated using your student ID

Instructions:

1. Derive $\frac{dZ(f)}{df}$
2. Reformulate the problem as a root-finding problem:

$$g(f) = \frac{dZ(f)}{df} - K = 0$$

3. Use the **Bisection Method** to solve for f^*
4. Use an interval $[0, C]$
5. Stop when:

$$|g(f)| < 10^{-5}$$

Student-Specific Parameters:

Let your student ID be ID :

- $seed = \text{int}(\text{hashlib.sha256}(\text{student_id.encode()}).\text{hexdigest}(), 16)\%1e8$
- $B = 0.15 + \frac{\text{int}(seed)\%10}{100}$
- $T^0 = 5 + (\text{int}(seed)\%10)$
- $K = T^0 + (\text{int}(seed)\%(T^0 * (1 + 5 * B) - T^0))$

IMPORTANT NUMBER

$x^* = \text{round}(f^*)$ and $y^* = \text{round}(Z(f^*))$. Compute:

$$R_1 = \text{int}(\text{hash}(\text{tuple}([x^*, y^*]))\%100000)$$

Problem 2: Golden Section Search for Optimal Step Size

This problem mimics the line search step used in traffic assignment algorithms such as FrankWolfe.

You are given a subset of links from the Sioux Falls network. For each link (i, j) , two flow values are provided:

- Current flow x_{ij}
- Auxiliary flow y_{ij}

Define the flow on each link as a function of λ :

$$f_{ij}(\lambda) = x_{ij} + \lambda(y_{ij} - x_{ij}), \quad \lambda \in [0, 1]$$

The objective function is the total system travel time:

$$Z(\lambda) = \sum_{(i,j)} f_{ij}(\lambda) \cdot T_{ij}(f_{ij}(\lambda))$$

where the travel time function follows the BPR form:

$$T_{ij}(f) = T_{ij}^0 \left(1 + B_{ij} \left(\frac{f}{C} \right)^4 \right), \quad C = 1000$$

Input Data:

For each student, a data file is provided. This file contains:

- A list of selected links (i, j)
- For each link:

- x_{ij} : current flow
- y_{ij} : auxiliary flow
- T_{ij}^0 : free-flow travel time
- B_{ij} : BPR parameter

Each student must use the dataset corresponding to their `student_id`.

Instructions:

1. Read the provided data file corresponding to your `student_id`
2. Implement the **Golden Section Search** algorithm
3. Minimize $Z(\lambda)$ over $\lambda \in [0, 1]$
4. Stop when:

$$|b - a| < 10^{-5}$$

IMPORTANT NUMBER

Let λ^* be the optimal value rounded to 6 decimal places and $y^* = \text{round}(Z(\lambda^*))$. Compute:

$$R_2 = \text{int}(\text{hash}(\text{tuple}([x_{g\text{olde}}, y_{g\text{olde}}])) \% 100000)$$

Submission for Part 3

Submit:

- Your implementation of:
 - Bisection Method.
 - Golden Section Search.
 - The code to solve problem 1 and problem 2 using the above methods.
- A short report including:
 - Derived formulas
 - Final values of f^* and λ^*
- A single text file containing IMPORTANT NUMBERS:
 - R_1
 - R_2